

London-Rabi response of a superconducting qubit

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Coherent manipulation of superconducting circuits for quantum computing necessitates a detailed understanding of their electromagnetic properties during operation. Two big ones are Meissner effect and Rabi oscillations. Despite the large body of research covering these two light-matter interactions, the literature lacks a self-consistent framework unifying them, which is important because both are at play in these important systems. By borrowing language from the cold atoms community, we provide a minimal model capable of incorporating both effects and demonstrate its predictive power by simulating their interplay in a superconducting charge qubit periodically driven by a dipole source. Future directions should consider the integration of more microscopic details and more realistic geometries.

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I. INTRODUCTION

Superconductors have long captivated the condensed matter physics community for their exotic material properties, including but not limited to vanishing electrical resistance, persistent currents, and the expulsion of magnetic fields. Despite the predictive power of the BCS theory, phenomenological models put forth by the London brothers and GL theory remain in widespread use for their ability to capture the electromagnetic properties of superconducting circuits with relatively few dynamical degrees of freedom.

In tandem, the atomic, molecular, and optical physics community developed a number of models for atoms... Rabi oscillations...

We provide a unified model. In section II, we outline the model. In section III, we present a simulation of the interplay between Rabi oscillations and the Meissner effect in a superconducting charge qubit. In section IV, we discuss next steps.

II. MODEL

A. Field theory

The fundamental assumption of our model is that the Cooper pairs can be treated as spinless bosons of charge q and mass m and thus described by the field Ψ coupled to a $U(1)$ gauge field $(V, \mathbf{A})$. Considering an external static charge background and an external current ρ_n and $\mathbf{J}_n$, respectively, gives us the following non-relativistic Lagrangian

$$\mathcal{L} = \Psi^\dagger \left[i\hbar D_t - \frac{\hbar^2}{2m} \mathbf{D}^2 \right] \Psi + \frac{\epsilon_0}{2} \mathbf{E}^2 - \frac{1}{2\mu_0} \mathbf{B}^2 - \rho_n V + \mathbf{J}_n \cdot \mathbf{A} \quad (1)$$

Coupling to the gauge field is given by the covariant derivatives

$$D_t \equiv \partial_t + (iq/\hbar)V \quad (2a)$$

$$\mathbf{D} \equiv \nabla - (iq/\hbar)\mathbf{A} \quad (2b)$$

with the electric and magnetic fields defined in the usual way

$$\mathbf{E} = -\nabla V - \dot{\mathbf{A}} \quad (3a)$$

$$\mathbf{B} = \nabla \times \mathbf{A}. \quad (3b)$$

The system's local $U(1)$ symmetry gives rise to a Noether current of the form

$$\rho_s \equiv q\Psi^\dagger \Psi \quad (4a)$$

$$\mathbf{J}_s \equiv \frac{q\hbar}{m} \text{Im} [\Psi^\dagger \mathbf{D}\Psi]. \quad (4b)$$

The total charge density $\rho = \rho_n + \rho_s$ and current density $\mathbf{J} = \mathbf{J}_n + \mathbf{J}_s$ are given by the sum of the normal and superconducting components. Performing a Legendre transformation in the Coulomb gauge gives the Hamiltonian

$$H = \int \left[\frac{\hbar^2}{2m} (\mathbf{D}\Psi)^\dagger \cdot \mathbf{D}\Psi + \frac{\epsilon_0}{2} \mathbf{E}_\perp^2 + \frac{1}{2\mu_0} \mathbf{B}^2 + \frac{1}{2} \rho (G * \rho) + \rho_n V - \mathbf{J}_n \cdot \mathbf{A}_\perp \right] d^3x \quad (5)$$

Vector operators $\mathbf{E}_\perp$ and $\mathbf{A}_\perp$ are the divergence free components of $\mathbf{E}$ and $\mathbf{A}$, respectively, and $G(\mathbf{x} - \mathbf{x}')$ satisfies $-\nabla^2 G(\mathbf{x} - \mathbf{x}') = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$. Here, we state the canonical commutation relations

$$[\Psi(\mathbf{x}), \Psi^\dagger(\mathbf{x}')] = \delta^{(3)}(\mathbf{x} - \mathbf{x}') \quad (6a)$$

$$[\mathbf{A}_{\perp,i}(\mathbf{x}), -\epsilon_0 \mathbf{E}_{\perp,j}(\mathbf{x}')] = i\hbar \delta_{ij} \delta_\perp^{(3)}(\mathbf{x} - \mathbf{x}'). \quad (6b)$$

We model the dynamics using a Dirac-Frenkel Action

$$S = \int dt \langle \psi | H - i\hbar \partial_t | \psi \rangle \quad (7)$$

with a product state ansatz for the light and matter

$$|\psi\rangle = |\psi_L\rangle \otimes |\psi_M\rangle. \quad (8)$$

We assume the light may be described by a coherent state

$$|\psi_L\rangle = e^{-\epsilon_0 \frac{i}{\hbar} \int d^3x [(\mathbf{E}_\perp) \cdot \mathbf{A}_\perp + (\mathbf{A}_\perp) \cdot \mathbf{E}_\perp]} |\text{vac}\rangle \quad (9)$$

where the expectation values are defined via $\langle \cdot \rangle \equiv \langle \psi | \cdot | \psi \rangle$. With a Josephson junction geometry in mind, we assume the matter may be described within a two mode approximation

$$|\psi_M\rangle = \sum_{n_L + n_R = N} c_{n_L, n_R} \frac{(b_L^\dagger)^{n_L} (b_R^\dagger)^{n_R}}{\sqrt{n_L! n_R!}} |\text{vac}\rangle \quad (10)$$

with $c_{n_L, n_R} \in \mathbb{C}$ an undetermined set of coefficients and $n_L, n_R \in \mathbb{Z}_{\geq 0}$ the occupation numbers of the left and right modes, respectively. In the above definition, we have introduced the ladder operators

$$b_j(t) = \int \phi_j^*(\mathbf{x}) \Psi(\mathbf{x}, t) d^3x, \quad (11)$$

consistent with an expansion of the field operator

$$\Psi(\mathbf{x}, t) = \sum_{j=L, R} \phi_j(\mathbf{x}) b_j(t) \quad (12)$$

over an incomplete basis of orthonormal modes $\{\phi_L, \phi_R\}$ and exploited the $U(1)$ symmetry of the model to fix the total particle number $N = b_L^\dagger b_L + b_R^\dagger b_R = \int \Psi^\dagger \Psi d^3x$. We note that the ladder operators satisfy the bosonic algebra $[b_i, b_j^\dagger] = \delta_{ij}$. We introduce the number imbalance operator $n \equiv (b_L^\dagger b_L - b_R^\dagger b_R)/2$, and express the matter ansatz in its eigenbasis $\{|n\rangle\}$ via $|\psi_M\rangle = \sum_n c_n |n\rangle$ with $n \in [-N/2, -N/2 + 1, \dots, N/2]$.

Using these ansatzes for the light and matter, we may now take variations of the Dirac-Frenkel Action with respect to free parameters $\mathbf{A}_{\perp \text{cl}}, \mathbf{E}_{\perp \text{cl}}$, and c_n . Assuming the modes ϕ_L and ϕ_R are fixed, we obtain the equations of motion (EOM), resembling an effective Maxwell and Schrodinger Equation coupled to each other via $\mathbf{A}_{\perp \text{cl}}$:

$$\square(\mathbf{A}_\perp) = \mu_0 \Delta_\perp * \langle \mathbf{J} \rangle \quad (13a)$$

$$i\hbar \dot{c}_n = \sum_{n'} \mathcal{H}_{n, n'}(t) c_{n'} \quad (13b)$$

For transverse vector delta function $\Delta_\perp$, that will return the divergence free component of a real vector field when convolved with it. The matrix element $\mathcal{H}_{n, n'}$ is given by $\mathcal{H}_{n, n'} = \langle n | H | n' \rangle$, for time-dependent Hamiltonian

$$H = \sum_{i, j \in \{L, R\}} \epsilon_{ij}(\mathbf{A}) b_i^\dagger b_j + \sum_{i, j, k, l \in \{L, R\}} U_{ijkl}(\mathbf{A}) b_i^\dagger b_j^\dagger b_k b_l \quad (14)$$

with coefficients given by the following mode-dependent integrals:

$$\epsilon_{ij}(\langle \mathbf{A}_\perp \rangle) \equiv \int \phi_i^* \left(\frac{1}{2m} \left(\frac{\hbar}{i} \nabla - q \langle \mathbf{A}_\perp \rangle \right)^2 + U_{\text{ext}} \right) \phi_j d^3x \quad (15a)$$

$$U_{\text{ext}}(x) = -\frac{q^2 n_s}{\epsilon_0} G * \Theta \quad (15b)$$

$$U_{ijkl} \equiv \frac{q^2}{2\epsilon_0} \int \phi_i^* \phi_l (G * (\phi_j^* \phi_k)) d^3x \quad (15c)$$

B.

We will restrict our study to charge neutral systems with piecewise constant (“jellium”) charge backgrounds that have in the x direction. We can thus express $\rho = -qn_s \Theta$, for effective boson number density n_s and indicator function Θ that satisfies $\Theta(x, y, z) = \Theta(-x, y, z)$. To model a Josephson junction, we can define separated left and right “islands” of superconductor and corresponding indicator functions Θ_L and Θ_R such that $\Theta = \Theta_L + \Theta_R$ and $\Theta_L(-x, y, z) = \Theta_R(x, y, z)$. To further simplify our analysis, we will also assume that the two modes ϕ_L and ϕ_R have the same in the x direction, such that $\phi_L(x, y, z) = \phi_R(-x, y, z)$. By charge neutrality, we expect the shapes of corresponding left and right modes ϕ_L and ϕ_R to closely resemble the left and right indicator functions. Thus, for volume $V = \int \Theta d^2x$, we expect $(V/2)|\phi_L|^2 \approx \Theta_L$ and $(V/2)|\phi_R|^2 \approx \Theta_R$.

C. Electrostatics

Considering only the steady state solution of matter in a vacuum, we may set $|\psi_L\rangle = |\text{vac}\rangle$ and removing time dependence from all variables. This yields the following variational problem for the modes ϕ_L and ϕ_R , and the coefficients c_n , in terms of the matter Hamiltonian H_M :

$$\delta \langle \psi_M | H_M | \psi_M \rangle = 0 \quad (16a)$$

$$H_M = \frac{\hbar^2}{2m} \int \nabla \Psi^\dagger \cdot \nabla \Psi + \frac{1}{2} \rho G * \rho d^3x \quad (16b)$$

Where H_M in the two mode approximation reduces to the Hamiltonian in Equation 14, with $\mathbf{A}_{\perp, \text{cl}} = 0$. Defining effective spin operators that satisfy the $SU(2)$ commutation relations with total angular momentum $N/2$ [VYC: “Inverse Holstein-Primakoff” transform??] [TJM: This is not the Holstein-Primakoff transform but rather the “inverse” of the Schwinger boson representation]:

$$J_x \equiv \frac{1}{2} (b_L^\dagger b_R + b_R^\dagger b_L) \quad (17a)$$

$$J_y \equiv \frac{1}{2i} (b_L^\dagger b_R - b_R^\dagger b_L) \quad (17b)$$

$$J_z \equiv \frac{1}{2} (b_L^\dagger b_L - b_R^\dagger b_R) = \frac{1}{2} (n_L - n_R) = n \quad (17c)$$

the Hamiltonian can be expressed as

$$H = 2(\alpha' - \beta' - \gamma')J_z^2 + (\epsilon_{LR} + 2(N-1)\Delta')(J_+ + J_-) + \gamma'(J_+^2 + J_-^2) + (\alpha' + \beta' + \gamma')\frac{N^2}{2} + (\epsilon_{LL} - \alpha')N \quad (18)$$

For integrals:

$$\alpha' = U_{LLLL} = U_{RRRR} \quad (19a)$$

$$\beta' = U_{LRRL} = U_{RLLR} \quad (19b)$$

$$\gamma' = U_{LRLR} = U_{RLRL} = U_{LLRR} = U_{RRLL} \quad (19c)$$

$$\Delta' = U_{LLLR} = U_{RRRL} = \dots \quad (19d)$$

We introduce the phase operator φ , defined as the canonical conjugate of the charge imbalance operator, $[\varphi, n] = i$. It can be shown using $e^{\pm i\varphi}|n\rangle = |n \pm 1\rangle$ that:

$$J_{\pm}|n\rangle = \sqrt{(N/2)(N/2+1) - n(n+1)}e^{\pm i\varphi}|n\rangle \quad (20)$$

When N is small, the Josephson junction is in the so-called finite island regime and exhibits distinct behavior determined by the square root in Equation 20. However, to simplify our analysis, we will truncate the matter Hilbert space at a charge imbalance cutoff $n_{\max} < N$, such that $n \in [-n_{\max}/2, \dots, n_{\max}/2]$, under the assumption that the ground state coefficient distribution for c_n will be highly localized around zero charge imbalance, such that for all $|n| > n_{\max}/2$, $c_n \approx 0$.

We will assume that $N \gg n_{\max}$ such that we are in the infinite island regime, so $N \gg n$ for all n in Equation 20. In this limit, $J_{\pm} \approx \frac{N}{2}e^{\pm i\varphi}$. Additionally, we make the approximation that $\gamma' \approx 0$, if we expect the modes ϕ_L and ϕ_R to spatially resemble the respective left and right islands of charge background Θ , as discussed above. Under both of these approximations, the Hamiltonian reduces to that of an effective charge qubit with a zero point energy.

$$H = E_C n^2 - E_J \cos \varphi + \mathcal{E}_0 \quad (21)$$

With:

$$E_C = 2(\alpha' - \beta') \quad (22a)$$

$$E_J = -N(\epsilon_{LR} + 2(N-1)\Delta') \quad (22b)$$

$$\mathcal{E}_0 = (\alpha' + \beta')\frac{N^2}{2} + (\epsilon_{LL} - \alpha')N \quad (22c)$$

D. Dynamics

We now consider the case of nontrivial, time varying $\mathbf{A}_{\perp\text{cl}}$, sourced by a sinusoidal external current $\mathbf{j}$ of a single frequency ω_d that is turned on at $t = 0$.

We will fix the modes ϕ_L and ϕ_R to obtain the system dynamics from Equations 13a and 13b. To linearize

Equation 13a, we will make the approximation that the expected Cooper pair density $\langle \psi_M | \Psi^\dagger \Psi | \psi_M \rangle$ does not vary in time.

[TJM: This is probably the most delicate part of the entire derivation. I don't think it relies on sinusoidal perturbations. We should frame it an instance of the Born approximation... really just first-order truncation of Dyson series, which can be generalized by going to second order, etc. You should paste the Dyson series equation for expectations of time-evolved operators with strict equality. The Taylor series will be in time t . Then write $\approx$ the operator at time 0, so really I think it's a zeroth order approximation...] We will also make the approximation that the first term of the supercurrent in Equation 4b can be dropped assuming little spatial overlap of the modes and symmetric distribution of the coefficients in the charge imbalance basis, as:

$$\begin{aligned} & \langle \psi_M | \nabla \Psi^\dagger \Psi - \Psi^\dagger \nabla \Psi | \psi_M \rangle \\ &= ((\nabla \phi_L) \phi_R - (\nabla \phi_R) \phi_L) \langle \psi_M | b_L^\dagger b_R - b_R^\dagger b_L | \psi_M \rangle \end{aligned} \quad (23)$$

Under these approximations, the EOM for the classical vector potential $\mathbf{A}_{\perp\text{cl}}$ for the divergence-free source $\mathbf{j}_{\perp\text{n}} = \Delta_{\perp} * \mathbf{j}_{\text{n}}$ is

$$\square \mathbf{A}_{\perp\text{cl}} + \frac{\mu_0 q^2}{m} \langle \psi_M | \Psi^\dagger \Psi | \psi_M \rangle \mathbf{A}_{\perp\text{cl}} = \mu_0 \mathbf{j}_{\perp\text{n}} \quad (24)$$

[TJM: The currents should be lumped together and in fact I think the above equation is slightly wrong because you need to take the transverse part of second term on the LHS above...

$$\square \mathbf{A}_{\perp\text{cl}} = \mu_0 \Delta_{\perp} * \mathbf{j}_{\Sigma} \quad (25)$$

with $\mathbf{j}_{\Sigma}$ the total current (sum of super and normal).] For time-independent $\langle \psi_M | \Psi^\dagger \Psi | \psi_M \rangle$. For this linear PDE that is harmonically driven, we can make the ansatz

$$\mathbf{j}_{\perp}(\mathbf{x}, t) = \mathbf{j}_{\perp,0}(\mathbf{x}) \sin(\omega_d t) \quad (26a)$$

$$\mathbf{A}_{\perp\text{cl}}(\mathbf{x}, t) = \mathbf{A}_{\perp\text{cl},0}(\mathbf{x}) \sin(\omega_d t) \quad (26b)$$

Where deep in the bulk of the superconductor, we will assume there is no external current source from inside the superconductor $\mathbf{j}_{\perp} = 0$, and that superconducting charge density exact cancels the ionic charge background $\langle \psi_M | \Psi^\dagger \Psi | \psi_M \rangle = n_s$. Thus, in the bulk:

$$\nabla^2 \mathbf{A}_{\perp\text{cl},0} = \left(\frac{1}{\lambda_L^2} - \frac{\omega_d^2}{c^2} \right) \mathbf{A}_{\perp\text{cl},0} \quad (27a)$$

$$\lambda_L = \sqrt{\frac{m}{\mu_0 n_s q^2}} \quad (27b)$$

Where λ_L is the steady state London penetration depth, and Equation 27a reduces to the second London Equation in the steady state for $\omega_d = 0$. The effective London penetration depth is, in the two mode approximation:

$$\lambda_{L,\text{eff}} = \sqrt{\frac{c^2}{\omega_p^2 - \omega_d^2}} \quad (28)$$

For plasma frequency of the superconductor $\omega_p = \frac{c}{\lambda_L}$. Therefore, the external electromagnetic drive frequency must be lower than the plasma frequency for Meissner screening. Given $\mathbf{A}_{\perp,cl}$, the Hamiltonian for the time evolution of the coefficients c_n :

$$H(t) = H_M + H'(t) \quad (29a)$$

$$H'(t) = \frac{1}{2m} \sum_{\{i,j\} \in \{L,R\}} b_i^\dagger b_j \int \phi_i^* (iq\hbar (\nabla \mathbf{A}_{\perp,cl} + \mathbf{A}_{\perp,cl} \nabla) + q^2 \mathbf{A}_{\perp,cl}^2) \phi_j d^3x \quad (29b)$$

For small enough magnitude of external drive the terms linear $\mathbf{A}_{\perp,cl}$ will dominate the quadratic (second harmonic) term, thus in this regime, we expect this time-dependent perturbation to drive Rabi oscillations of a two-level system (TLS) of frequency ω_d . For the discrete Hamiltonian $\mathcal{H} = \sum_{n,n'=-N/2}^{N/2} \mathcal{H}_{n,n'} |n\rangle \langle n'|$, the ground and first excited states in the charge imbalance basis can be determined from E_J and E_C , and thus coefficients $c_n(t)$ can be projected onto $|0\rangle$ and $|1\rangle$ at all times t .

III. NUMERICAL SIMULATIONS

A. Energy minimization for initial condition

We consider a $L_x \times L_y \times L_z$ conducting cavity in 3D and two $l_x \times l_y \times l_z$ superconducting islands separated by a distance d . To find the modes ϕ_L, ϕ_R , and ground state coefficients c_n , we discretize the finite domain of the cavity with a symplectic mesh, and evaluate all integrals over the modes using the Finite Element Method (FEM). We used quadratic scalar nodal basis functions to represent scalar functions ϕ_L and ϕ_R with a mesh that is finer around the edges of the superconducting

islands. We impose Dirichlet boundary conditions on the modes and Green's functions at the boundaries of the cavity.

Figure 1 shows the numerical solutions for ϕ_L , ϕ_R , and c_n , minimizing the ground state expectation of the matter Hamiltonian, $\langle H_M \rangle$. The modes are very localized around their corresponding island, as expected, although they are slightly skewed toward each other. When the two superconducting islands are separated by a greater distance d , the coefficient distribution becomes more peaked in the charge imbalance basis, as there is greater energetic cost of Cooper pair tunneling.

B. London-Rabi response

Given ω_d , Equation 27a is numerically solved in its weak form with FEM using a Nedelec basis to obtain $\mathbf{A}_{\perp,cl,0}$, and thus $\mathbf{A}_{\perp,cl}$. Then, using the steady state solution for the modes ϕ_L and ϕ_R and ground state coefficients c_n as an initial condition for the system at $t = 0$, we fix the modes and evolve the coefficients using Equation 13b, numerically evaluating $\mathcal{H}_{nn'}(t)$ with the solutions for $\mathbf{A}_{\perp,cl}$, ϕ_L and ϕ_R .

Figure 2 shows the numerical solutions for $\mathbf{A}_{\perp,cl,0}$ and the projection of the matter state $|\psi_M\rangle$ onto the $|1\rangle$ state over time. A 1D cross section is fit to an exponential curve, and is compared to the effective London penetration depth derived in Equation 28. [VYC: Do we want to show a single solution, if so the same as the first figure, or different, and in 2D or 3D? Try for 3D.]

IV. OUTLOOK

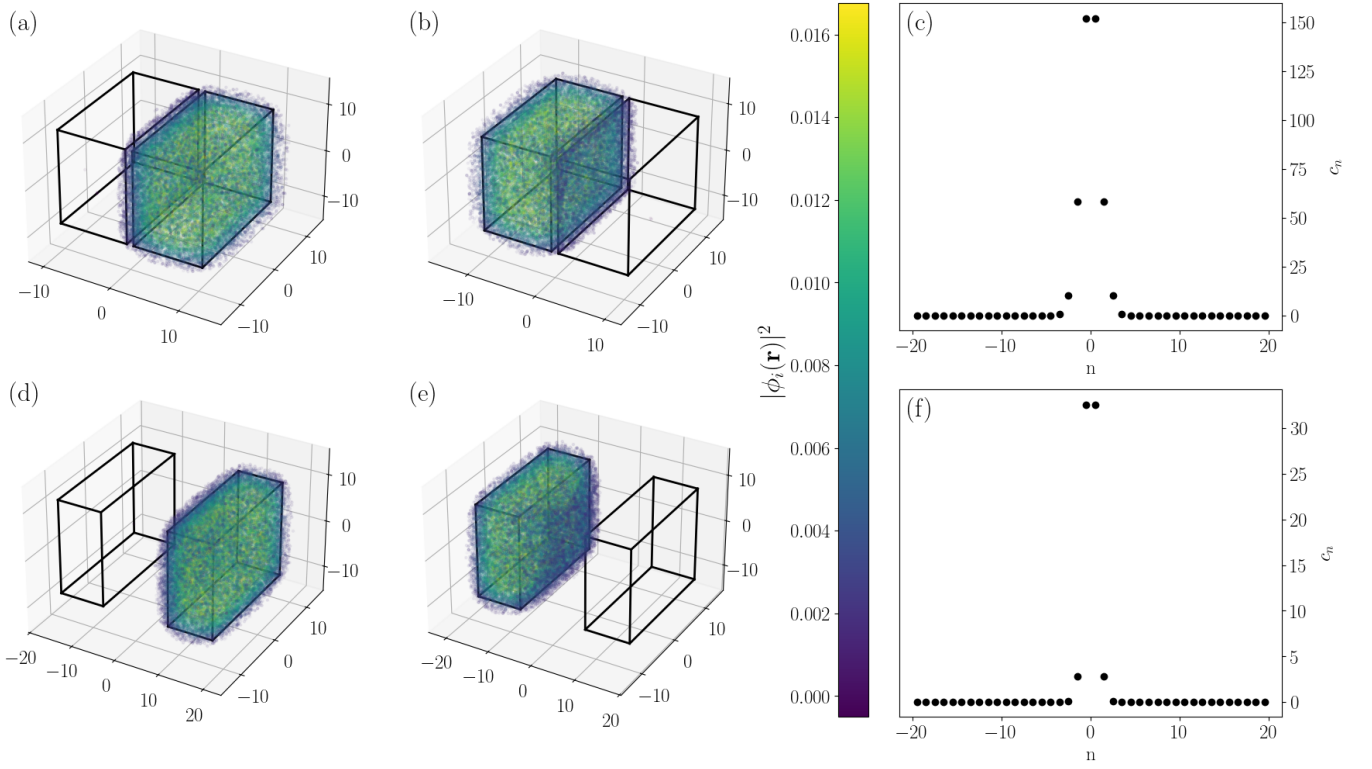


FIG. 1. Numerical solutions of the ground state optimization problem for two rectangular prisms, with dimensions $10 \times 20 \times 20$ in the x , y , and z directions, respectively, with separation of (a)–(c) 1, and (d)–(f) 15. The first column (a) and (d) show the left spatial mode $\phi_L(x)$, and the second column (b) and (e) show the right spatial mode $\phi_R(x)$. The third column (c) and (f) show the squared magnitude of the normalized coefficients with respect to the charge imbalance n . All length scales are in units of healing length ξ . The plots of the mode profiles were generated by sampling values at various points in 3D at random, ignoring any values under a threshold of 0.0025.

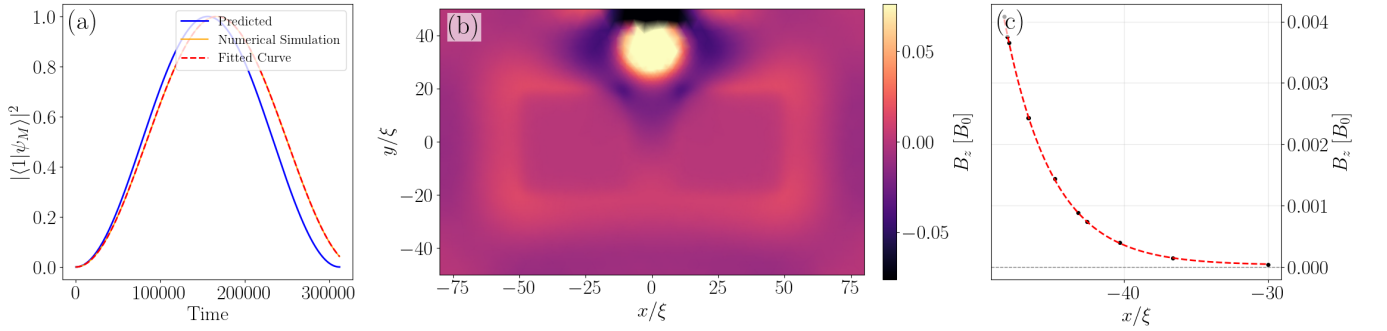


FIG. 2. London and Rabi responses of a superconducting circuit. (a) Rabi Oscillations, showing probability of measuring first excited state $|\langle 1 | \psi_M \rangle|^2$ over time. (b) 2D Heatmap showing $B_z = [\nabla \times \mathbf{A}_{\perp \text{cl}, 0}]_z$, in units of $B_0 = \frac{\hbar}{q\xi^2}$ (c) London penetration fit to a 1D cross section, with B_z values at all mesh points within a distance of 1ξ to $y = 0$.